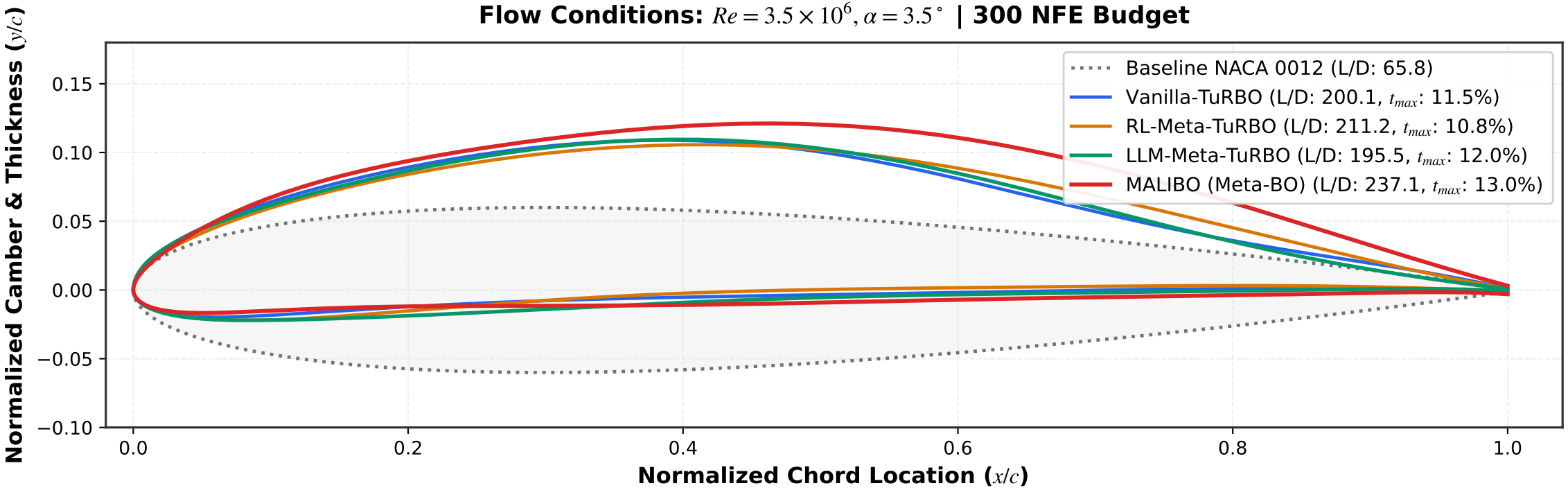
Optimized Airfoil Geometry Profiles Comparison (Representative Median Runs)
Flow Conditions: $Re = 3.5 \times 10^6, \alpha = 3.5^\circ$ | 300 NFE Budget
Baseline NACA 0012 (L/D: 65.8)
Vanilla-TuRBO (L/D: 200.1, t_{max}: 11.5%)
RL-Meta-TuRBO (L/D: 211.2, t_{max}: 10.8%)
LLM-Meta-TuRBO (L/D: 195.5, t_{max}: 12.0%)
MALIBO (Meta-BO) (L/D: 237.1, t_{max}: 13.0%)
Normalized Camber & Thickness (y/c)
0.15
0.10
0.05
0.00
−0.05
−0.10
0.0
0.2
0.4
0.6
0.8
1.0
Normalized Chord Location (x/c)